HW1

1. $\hat{y}_{kNN} = \dfrac{y_1 + y_2 + y_3}{3}$

estimate 150:

150 → 65kg
168 → 78kg
171 → 80kg

$$\hat{y}_{150} = \frac{65+78+80}{3} \approx 74.33\text{ kg}$$

estimate 155:

150 → 65kg
168 → 78kg
171 → 80kg

$$\hat{y}_{155} = \frac{65+78+80}{3} \approx 74.33\text{ kg}$$

estimate 165:

168 → 78kg
171 → 80kg
178 → 83kg

$$\hat{y}_{165} = \frac{78+80+83}{3} \approx 80.33\text{ kg}$$

estimate 180:

178 → 83kg
182 → 80kg
191 → 100kg

$$\hat{y}_{180} = \frac{83+80+100}{3} \approx 87.67\text{ kg}$$

2. $\hat{y}_{kNN} = \dfrac{w_1 y_1 + w_2 y_2 + w_3 y_3}{w_1 + w_2 + w_3}$ $w_i = \dfrac{1}{d_i}$

150:
$|150-150| = 0$ $w_1 = \frac{1}{0}$
$|150-168| = 18$ $w_2 = \frac{1}{18}$
$|150-171| = 21$ $w_3 = \frac{1}{21}$

$\hat{y}_{150} = 65\text{kg}$

155:
$|155-150| = 5$ $w_1 = \frac{1}{5}$
$|155-168| = 13$ $w_2 = \frac{1}{13}$
$|155-171| = 16$ $w_3 = \frac{1}{16}$

$$\hat{y}_{155} = \frac{\frac{1}{5}\times 65 + \frac{1}{13}\times 78 + \frac{1}{16}\times 80}{\frac{1}{5}+\frac{1}{13}+\frac{1}{16}} \approx 70.71\text{ kg}$$

165: $|165-168| = 3 \quad w_1 = 1/3$

$|165-171| = 6 \quad w_2 = 1/6 \qquad \hat{y}_{165} = \dfrac{\frac{1}{3}\times 78 + \frac{1}{6}\times 80 + \frac{1}{13}\times 83}{\frac{1}{3}+\frac{1}{6}+\frac{1}{13}} \approx 79.24\,kg$

$|165-178| = 13 \quad w_3 = 1/13$

190: $|190-178| = 12 \quad w_1 = 1/12$

$|190-182| = 8 \quad w_2 = 1/8 \qquad \hat{y}_{190} = \dfrac{\frac{1}{12}\times 83 + \frac{1}{8}\times 80 + 1\times 100}{\frac{1}{12}+\frac{1}{8}+1} \approx 96.76\,kg$

$|190-191| = 1 \quad w_3 = 1$

3. $J(x) = x^TQx + d^Tx + c \qquad Q = Q^T$

$$[x_1\ x_2]\begin{bmatrix} a & c \\ c & b \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \begin{bmatrix} ax_1 + cx_2 \\ cx_1 + bx_2 \end{bmatrix}$$

$$x_1(ax_1 + cx_2) + x_2(cx_1 + bx_2)$$

$$ax_1^2 + cx_1x_2 + cx_1x_2 + bx_2^2$$

$\nabla_x J(x)$: $\quad x^TQx = \sum_{i=1}^{n}\sum_{j=1}^{n} q_{ij}x_ix_j$

$$\frac{\partial(x^TQx)}{\partial x_k} = \sum_{i=1}^{n} q_{ik}x_i + \sum_{j=1}^{n} q_{kj}x_j = Qx_k + Q^Tx_k = 2Qx_k$$

$$\frac{\partial(d^Tx)}{\partial x} = d \qquad \frac{\partial c}{\partial x} = 0$$

$$\Rightarrow \nabla_x J(x) = 2Qx + d$$

Hessian: $\dfrac{\partial(2Qx+d)}{\partial x} = 2Q$

$$H_{ij} = \frac{\partial^2 J}{\partial x_i \partial x_j} = 2q_{ij}$$

4. $X \in \mathbb{R}^{n\times(p+1)}$ $Y \in \mathbb{R}^n$ $\hat\beta = (X^TX)^{-1}X^Ty$

let test sample $x' \in \mathbb{R}^{1\times p}$ $\tilde{x} = [1, x'] \in \mathbb{R}^{1\times(p+1)}$

$\hat{y} = \tilde{x}\hat{\beta}$

$$\Rightarrow \hat{y} = \tilde{x}(X^TX)^{-1}X^Ty$$

let $a = \tilde{x}(X^TX)^{-1}X^T$ $\Rightarrow$ $\hat{y} = \sum_{i=1}^{n} a_i y_i$

This shows that the prediction is the average weight of the train lable. This weights are not determined by nearest neighbor distance. They depend on the linear model.

5.

We have $y \in \mathbb{R}^n$ $\hat{y} = X(X^TX)^{-1}X^Ty$

we want to get $\hat{y} = Xb$

$\Rightarrow$ let $b = (X^TX)^{-1}X^Ty$

$\because X = [1\ x_0\ x_1 \cdots] \in \mathbb{R}^{n\times(p+1)}$

$\therefore b \in \mathbb{R}^{p+1}$

$\Rightarrow \hat{y} = Xb$ $\hat{y} \in Col(X)$

6.

$RSS = (y_i - \hat{y}_i)^2 \Rightarrow RSS(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T(y - X\beta)$

$e = y - \hat{y}$ We want to show $e \perp Col(X) \Rightarrow X^Te = 0$

$$\nabla_\beta RSS(\beta) = -2X^T(y - X\beta) = 0$$

$$\Rightarrow X^T(y - X\beta) = 0$$

$$\Rightarrow X^T(y - \hat{y}) = 0$$

$X^Te = 0$ Therefore, e is orthogonal to every col of X